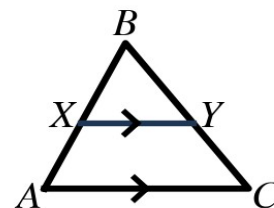


Speak Like A Mathematician:

What does 8.4 add to 8.2-8.3?



Proportionality Theorems

Triangle Proportionality Theorem: a line parallel to one side of a triangle divides the other two sides _____. **Converse:** if a line divides two sides proportionally, it is _____ to the third side.

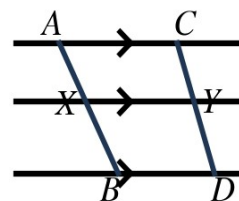
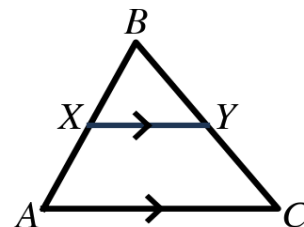
Three Parallel Lines: three parallel lines cut any two transversals into _____ segments.

Triangle Angle Bisector: an angle bisector of a triangle divides the opposite side into segments proportional to the other two _____.

Example 1 - Complete Each Proportion

In the triangle at the right, $\overline{XY} \parallel \overline{AC}$. Fill in the missing part.

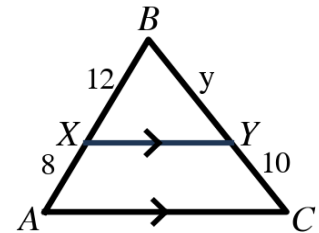
- $BX/XA = BY/\underline{\hspace{1cm}}$
- $XA/BX = YC/\underline{\hspace{1cm}}$
- $BX/BA = BY/\underline{\hspace{1cm}}$ (careful — whole sides now)
- In the three-parallel-lines figure at the right: $AX/XB = CY/\underline{\hspace{1cm}}$



Example 2 - Find the Missing Piece

In the triangle at the right, $\overline{XY} \parallel \overline{AC}$, $BX = 12$, $XA = 8$, $BY = y$, and $YC = 10$.

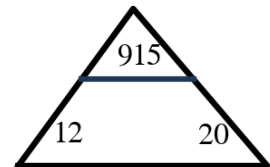
- Set up the proportion.
- Solve for y .



Example 3 - Is the Segment Parallel?

Use the Converse. The figures at the right show the segment pieces.

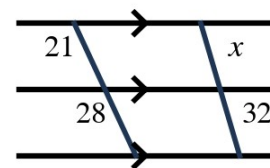
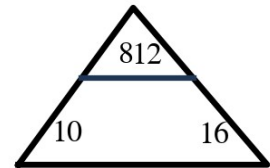
- First figure: pieces 9, 12 on one side and 15, 20 on the other.
- Second figure: pieces 8, 10 and 12, 16.



Example 4 - Three Parallel Lines

In the figure at the right, the three horizontal lines are parallel; the pieces measure 21, 28 on the left transversal and x , 32 on the right.

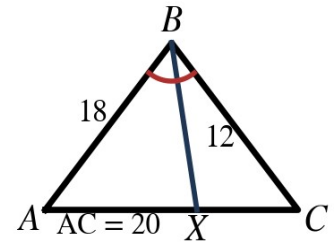
- Set up and solve.



Example 5 - The Angle Bisector Proportion

In $\triangle ABC$ at the right, $\overline{BX}$ bisects $\angle B$, $AB = 18$, $BC = 12$, and $AC = 20$.

- Write the proportion the bisector creates.
- Find AX and XC .
- Check the ratio.



Example 6 - Mixed Practice

Use the figures at the right.

- Three parallel lines: pieces 14, 21 on the left and DE, 27 on the right. Find DE.
- In $\triangle PQR$, $\overline{PS}$ bisects $\angle P$, $PQ = 15$, $PR = 10$, and $QR = 20$. Find QS and SR .
- In part b, which segment is longer, and why?

